

# Appendix for: Detecting and Investigating Complex Connectome Motifs in the Adult *Drosophila*

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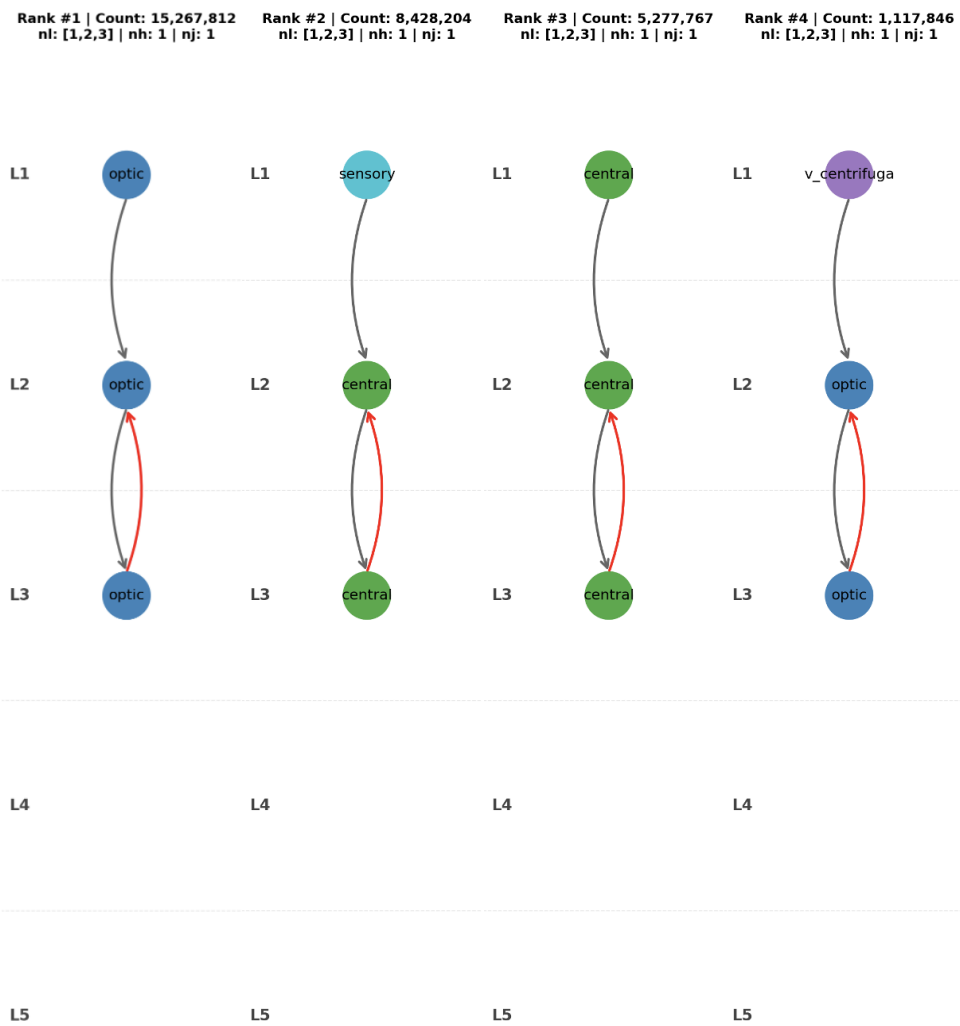
## Abstract

Understanding how cognitive functions emerge from the physical wiring of the brain is one of the main challenges in modern neuroscience. Recently, the full synaptic-resolution connectome of the adult *Drosophila melanogaster* (fruit fly) brain has been made available for computational examination. However, the nature of the raw connectome is a sparse, cyclic graph without any intrinsic coordinate system, making systematic analysis at circuit-level spanning across the entire depth of such brain structures a non-trivial algorithmic challenge. This work provides a computational pipeline that introduces a global hierarchical structure to the connectome via spectral graph embedding and data-driven level assignment. Based on this hierarchy, we propose a scalable parametric framework for motif discovery that abstracts the search space from individual neurons to hierarchies of functional areas. Using this pipeline on the full connectome of *Drosophila* we find several general architectural principles: the universal prevalence of feedforward-plus-local-feedback relays at all information processing tiers; a shift from optic-dominated recurrence in early stages to central-class integration at higher levels; the presence of large directed cycles in intermediate stages; and the coexistence of qualitatively different fast-and-slow circuit topologies. This framework is therefore general, parametric and directly applicable to other fully mapped connectomes, laying a reusable foundation for computational connectomics that goes beyond the fruit fly.

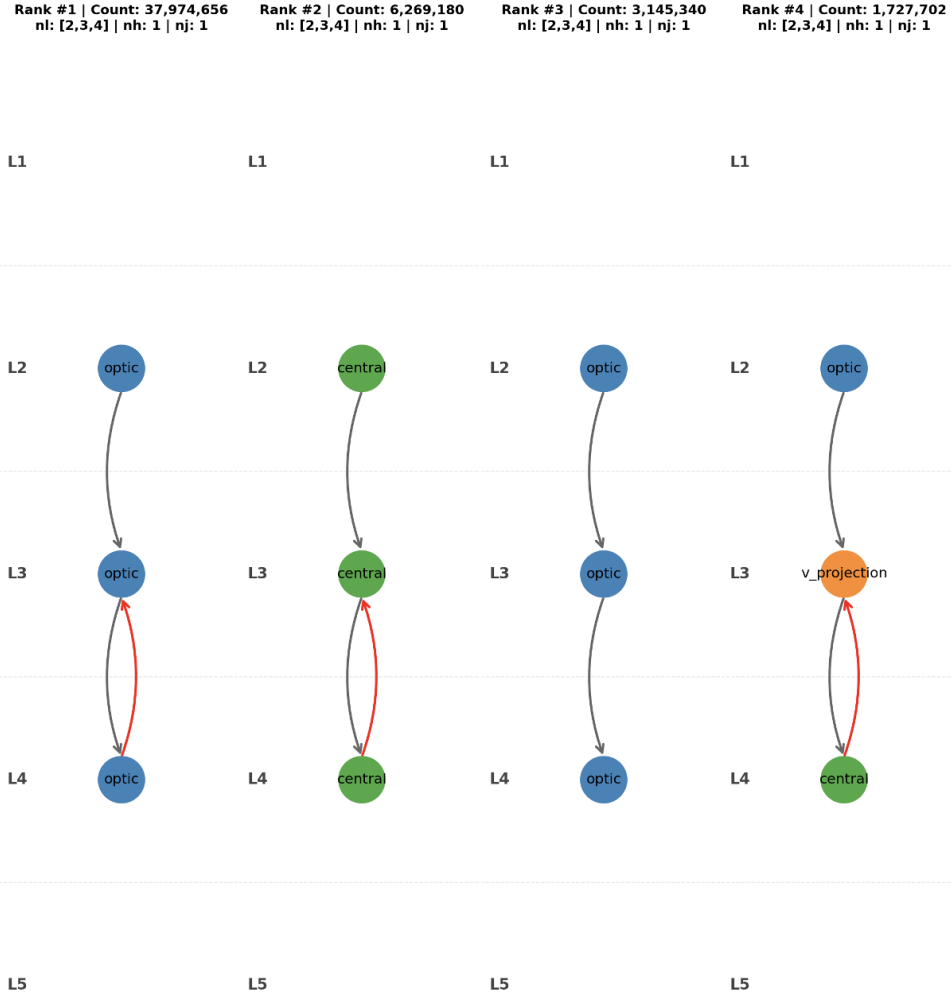
**Keywords:** *Drosophila melanogaster*; Connectome; Connectome Motifs; Complex Network Analysis;

## Results

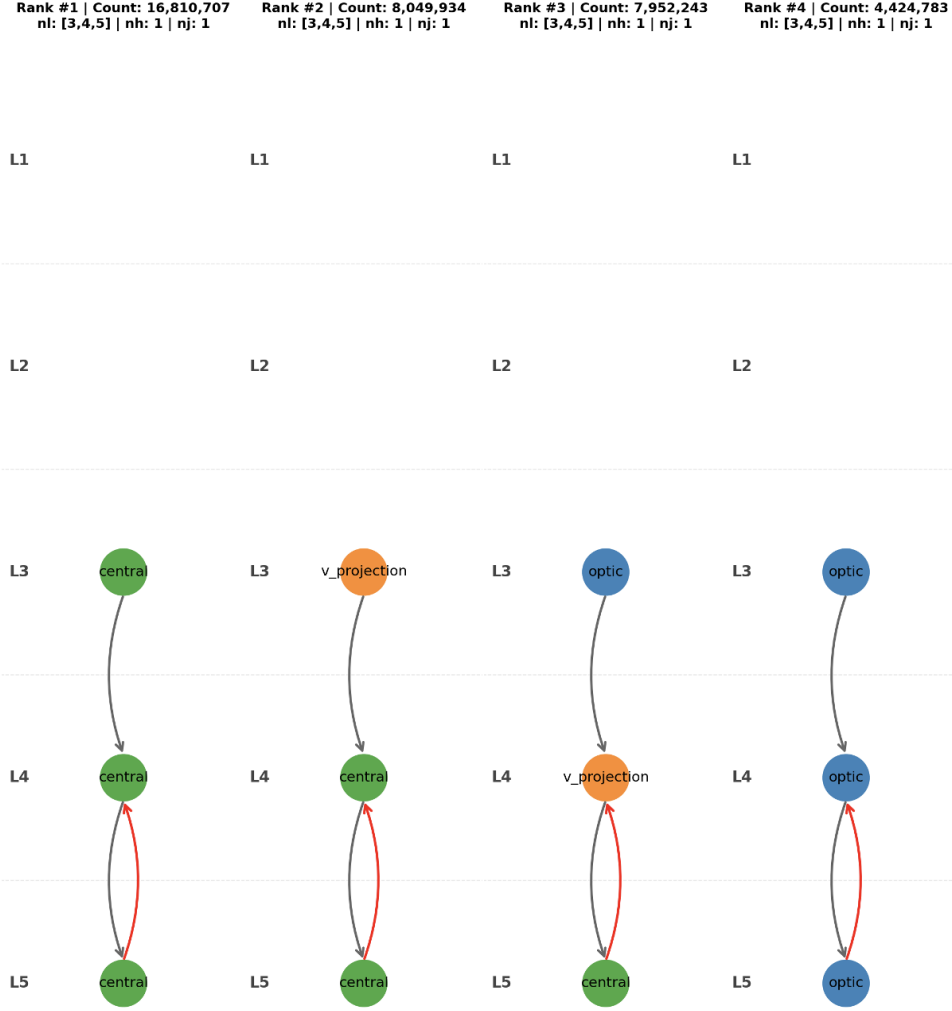
In this Appendix we report the whole set of results obtained from our motif discovery activity. Results are graphically shown as a list of figures. Each figure explains in the caption the values adopted for  $NL$ ,  $NH$ , and  $NJ$ . Moreover, in each figure the rank of each motif in its sub-group is reported, as well as its motif count. Gray arrows denote feedforward edges; red arrows denote feedback edges. Superclasses are consistently encoded with the same color across all figures to ensure visual alignment and immediate comparison. For details about the motif structures, parameters and representation please refer to the paper.



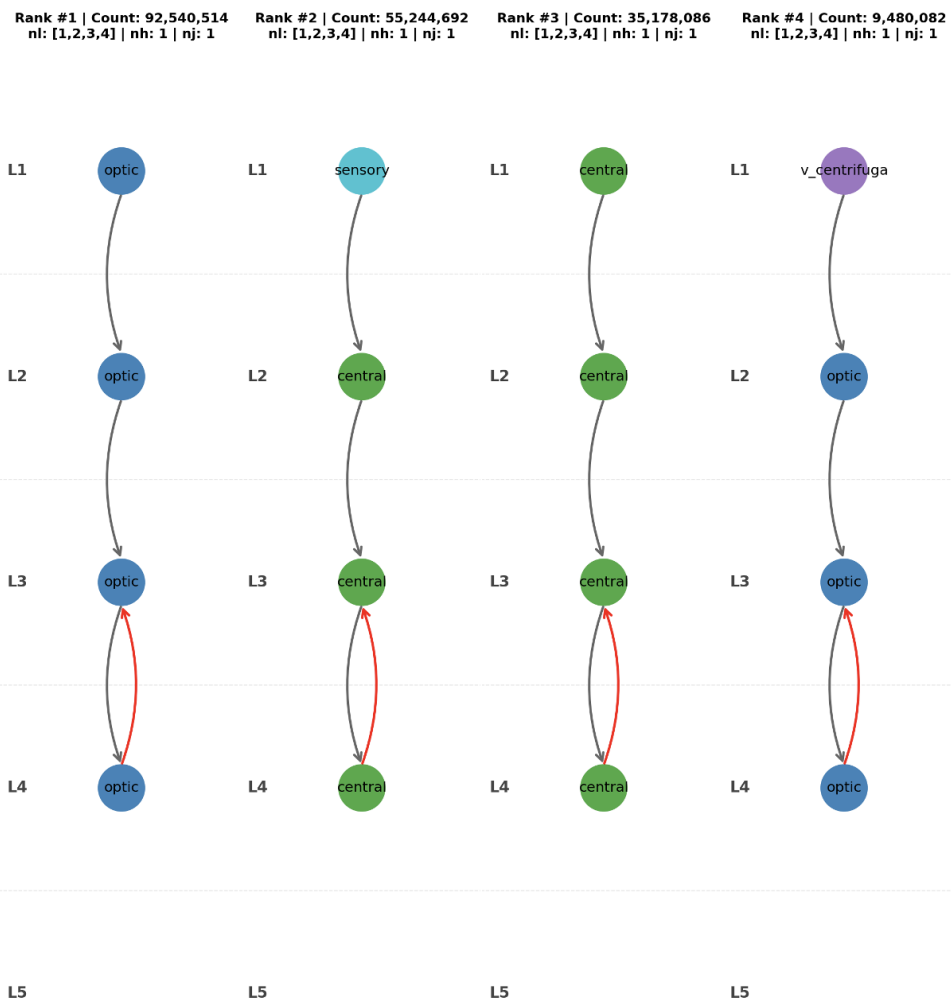
**Fig. 1** Top-4 ranked motifs discovered with parameters  $NL = \{1, 2, 3\}$ ,  $NH = 1$ ,  $NJ = 1$ .



**Fig. 2** Top-4 ranked motifs discovered with parameters  $NL = \{2, 3, 4\}$ ,  $NH = 1$ ,  $NJ = 1$ .

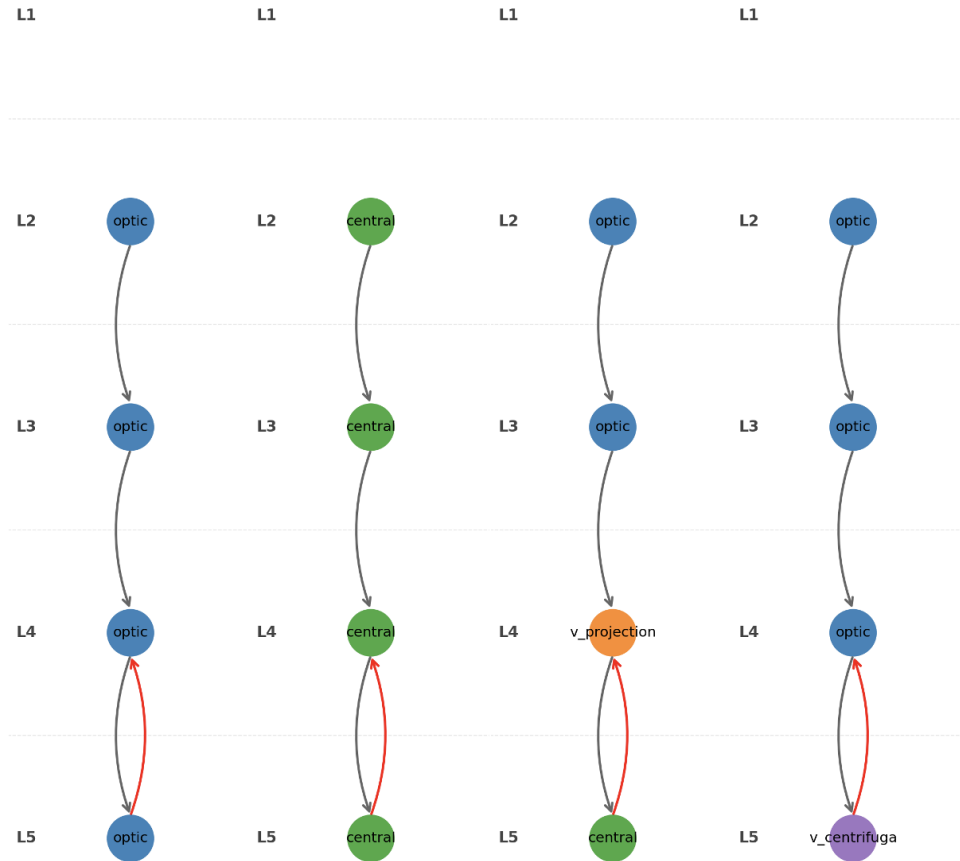


**Fig. 3** Top-4 ranked motifs discovered with parameters  $NL = \{3, 4, 5\}$ ,  $NH = 1$ ,  $NJ = 1$ .



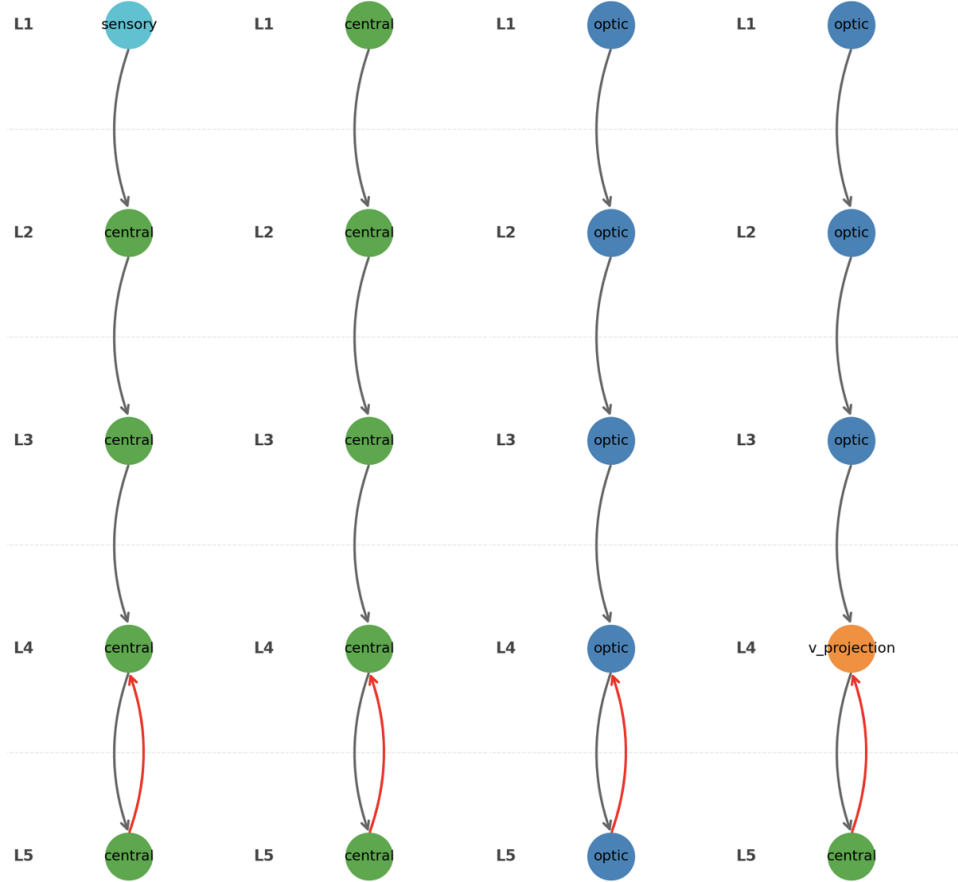
**Fig. 4** Top-4 ranked motifs discovered with parameters  $NL = \{1, 2, 3, 4\}$ ,  $NH = 1$ ,  $NJ = 1$ .

Rank #1 | Count: 361,842,766 | nl: [2,3,4,5] | nh: 1 | nj: 1    Rank #2 | Count: 171,209,328 | nl: [2,3,4,5] | nh: 1 | nj: 1    Rank #3 | Count: 122,965,931 | nl: [2,3,4,5] | nh: 1 | nj: 1    Rank #4 | Count: 90,106,043 | nl: [2,3,4,5] | nh: 1 | nj: 1



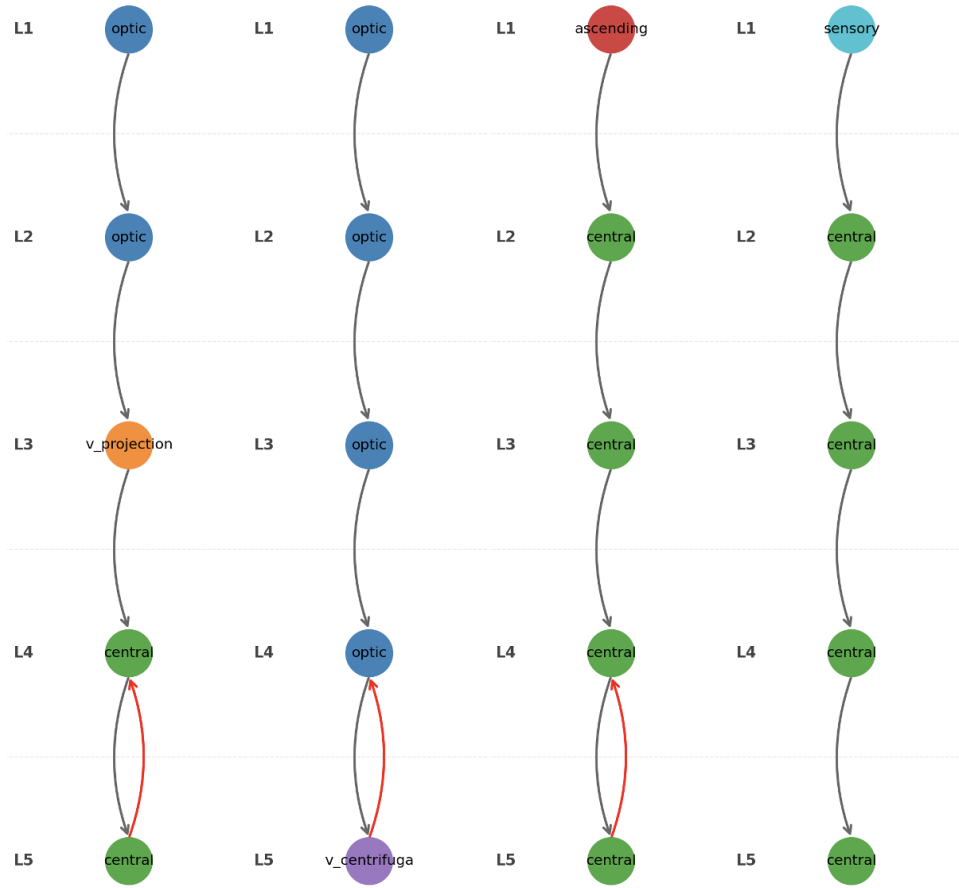
**Fig. 5** Top-4 ranked motifs discovered with parameters  $NL = \{2, 3, 4, 5\}$ ,  $NH = 1$ ,  $NJ = 1$ .

Rank #1 | Count: 971,032,327    Rank #2 | Count: 718,218,930    Rank #3 | Count: 503,158,268    Rank #4 | Count: 280,074,441  
nl: [1,2,3,4,5] | nh: 1 | nj: 1    nl: [1,2,3,4,5] | nh: 1 | nj: 1    nl: [1,2,3,4,5] | nh: 1 | nj: 1    nl: [1,2,3,4,5] | nh: 1 | nj: 1

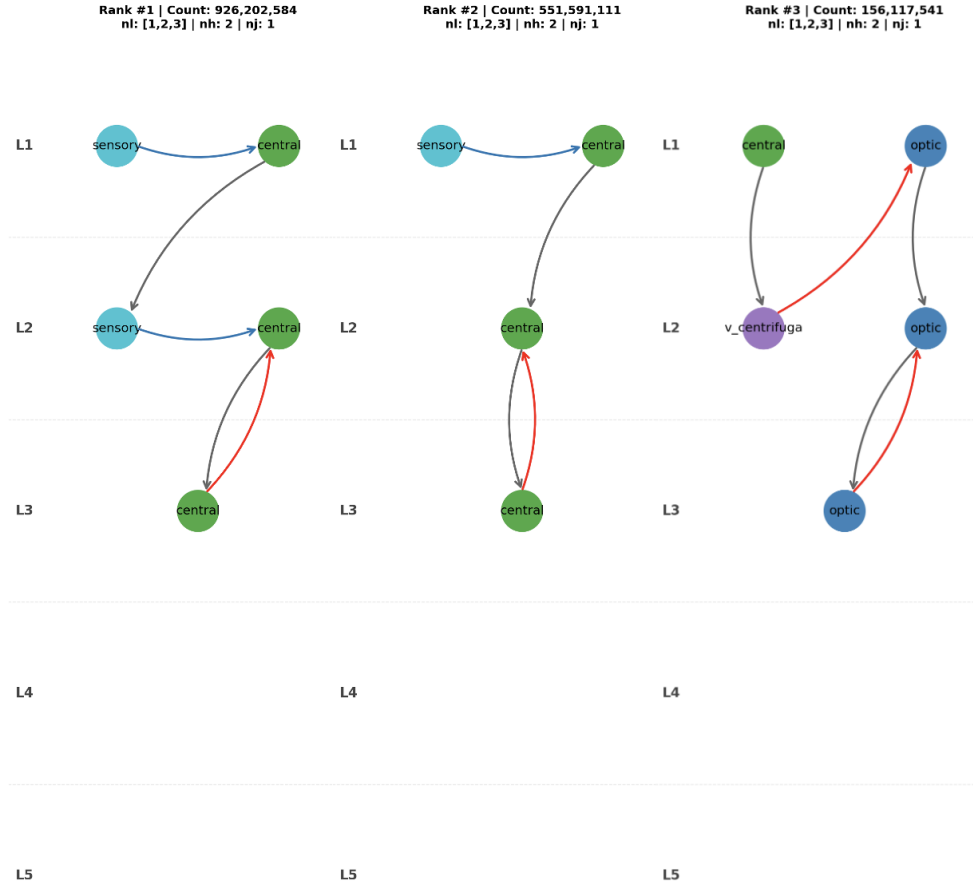


**Fig. 6** Top 1-4 ranked motifs discovered with parameters  $NL = \{1, 2, 3, 4, 5\}$ ,  $NH = 1$ ,  $NJ = 1$ .

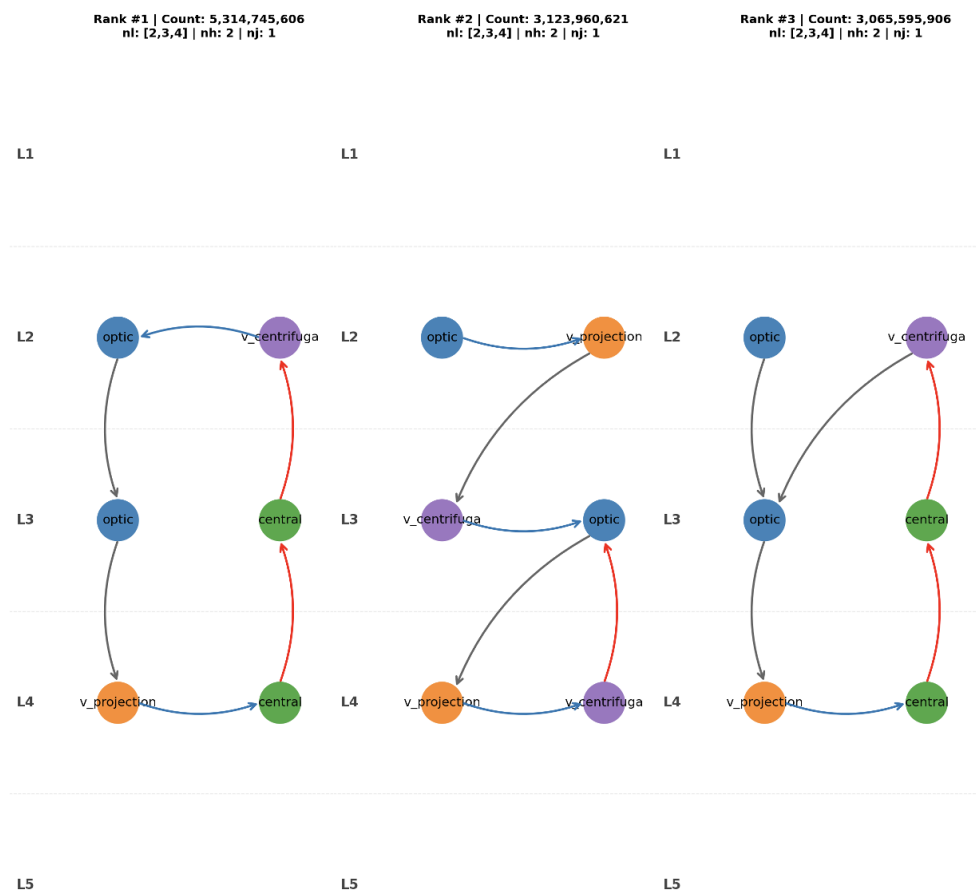
**Rank #5 | Count: 175,124,786**    **Rank #6 | Count: 107,369,613**    **Rank #7 | Count: 64,218,521**    **Rank #8 | Count: 47,300,566**  
 nl: [1,2,3,4,5] | nh: 1 | nj: 1    nl: [1,2,3,4,5] | nh: 1 | nj: 1    nl: [1,2,3,4,5] | nh: 1 | nj: 1    nl: [1,2,3,4,5] | nh: 1 | nj: 1



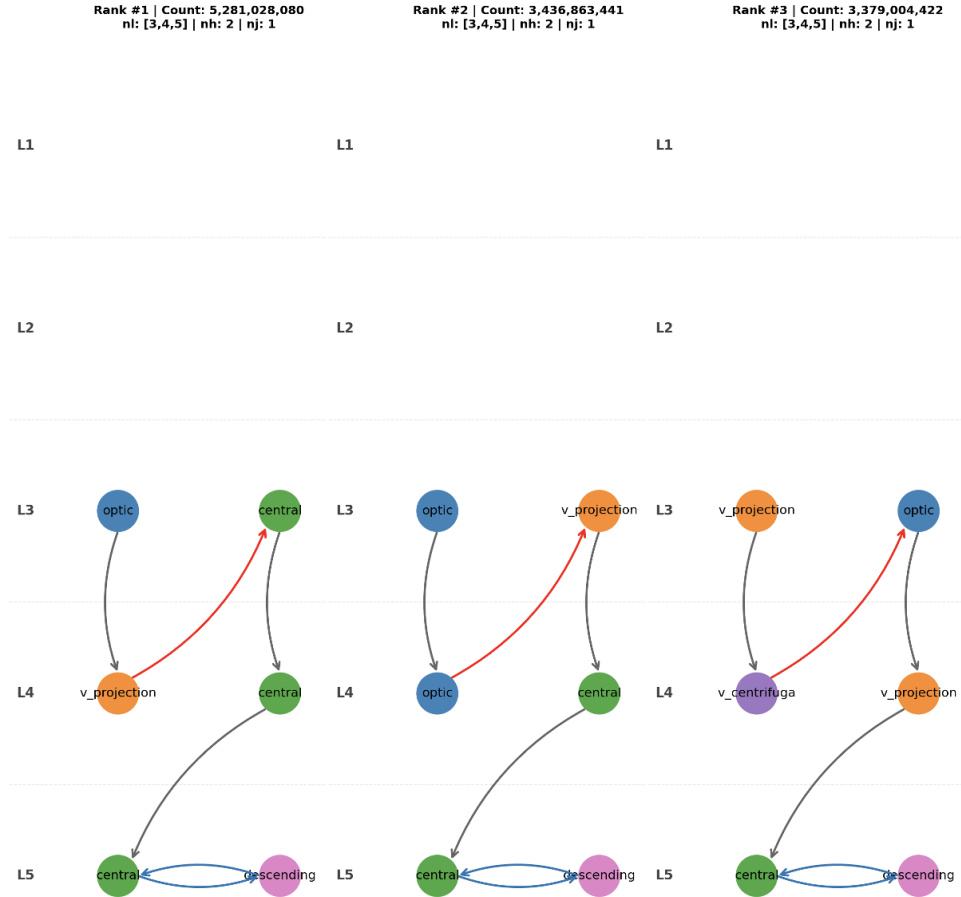
**Fig. 7** Top 5-8 ranked motifs discovered with parameters  $NL = \{1, 2, 3, 4, 5\}$ ,  $NH = 1$ ,  $NJ = 1$ .



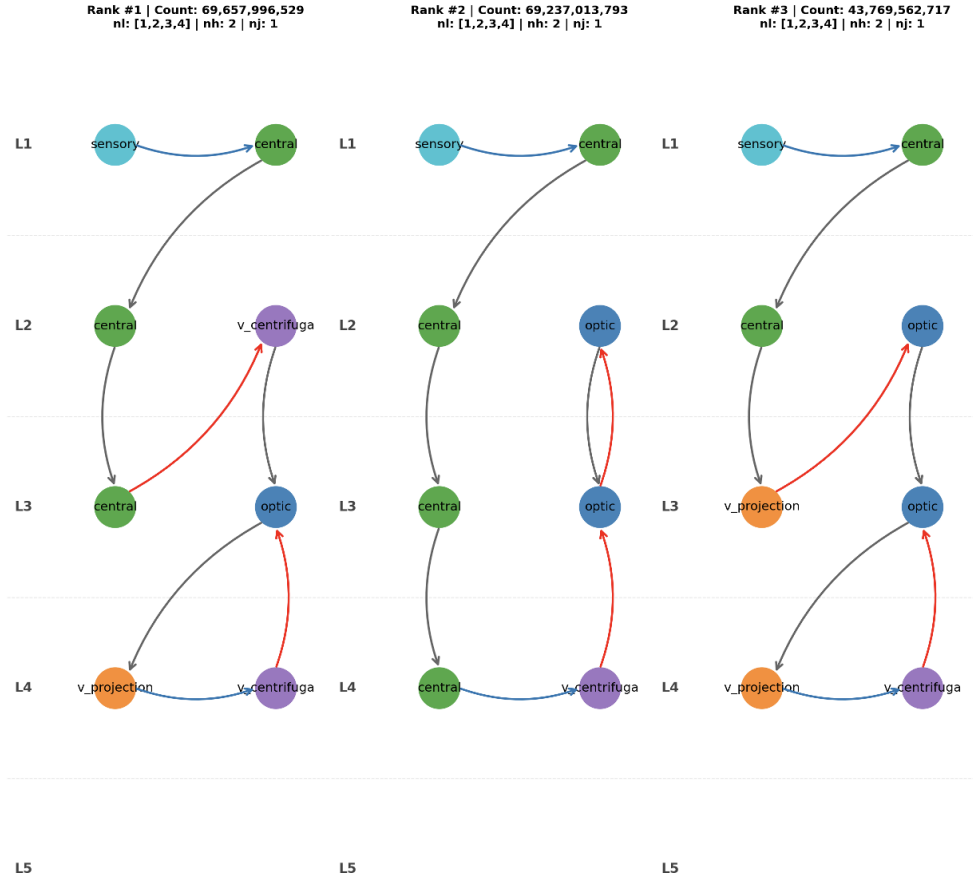
**Fig. 8** Top-3 ranked motifs discovered with parameters  $NL = \{1, 2, 3\}$ ,  $NH = 2$ ,  $NJ = 1$ .



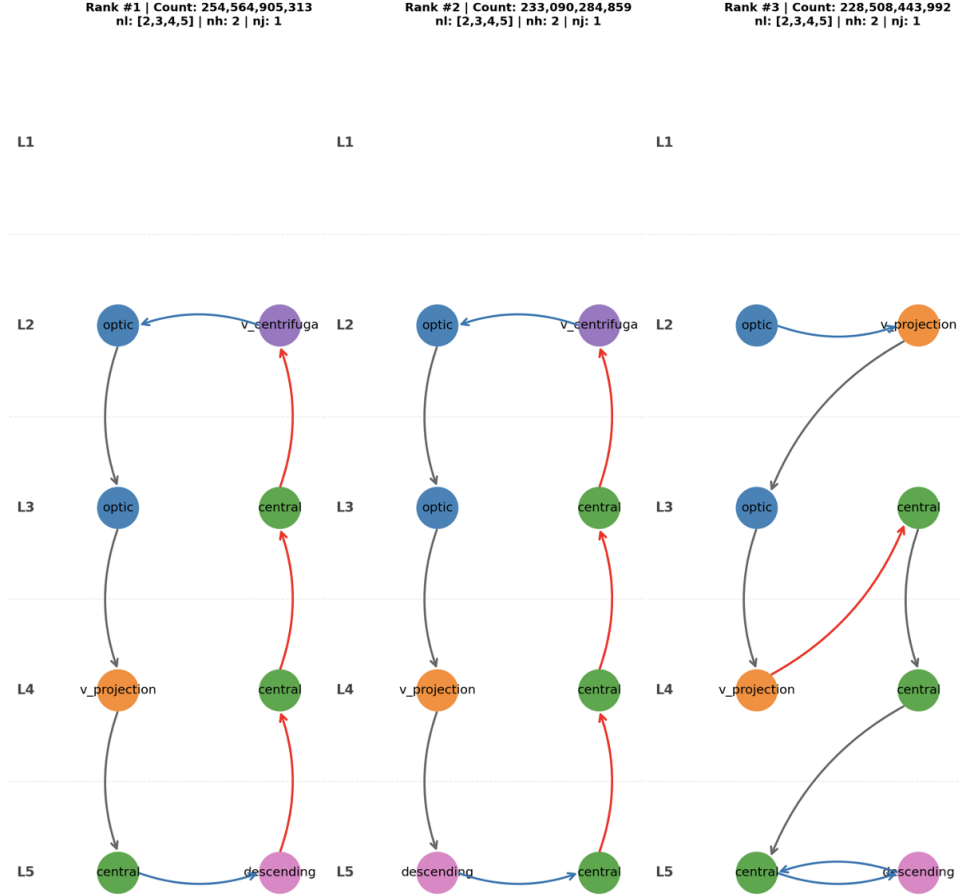
**Fig. 9** Top-3 ranked motifs discovered with parameters  $NL = \{2, 3, 4\}$ ,  $NH = 2$ ,  $NJ = 1$ .



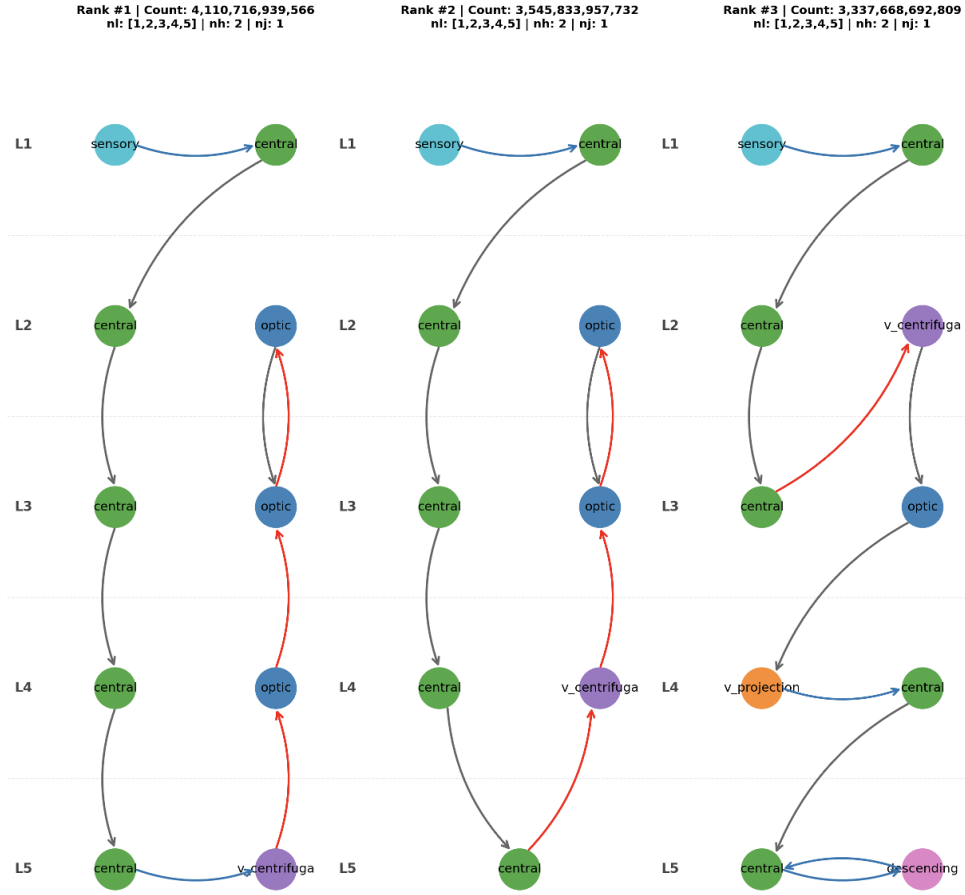
**Fig. 10** Top-3 ranked motifs discovered with parameters  $NL = \{3, 4, 5\}$ ,  $NH = 2$ ,  $NJ = 1$ .



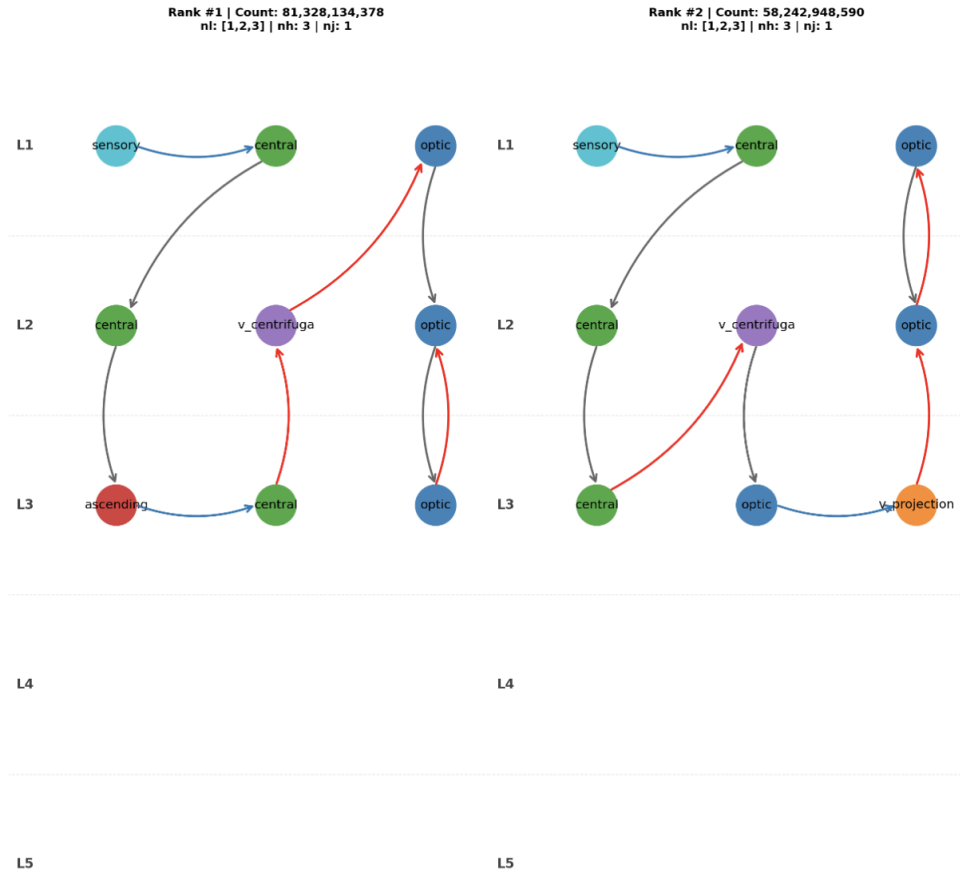
**Fig. 11** Top-3 ranked motifs discovered with parameters  $NL = \{1, 2, 3, 4\}$ ,  $NH = 2$ ,  $NJ = 1$ .



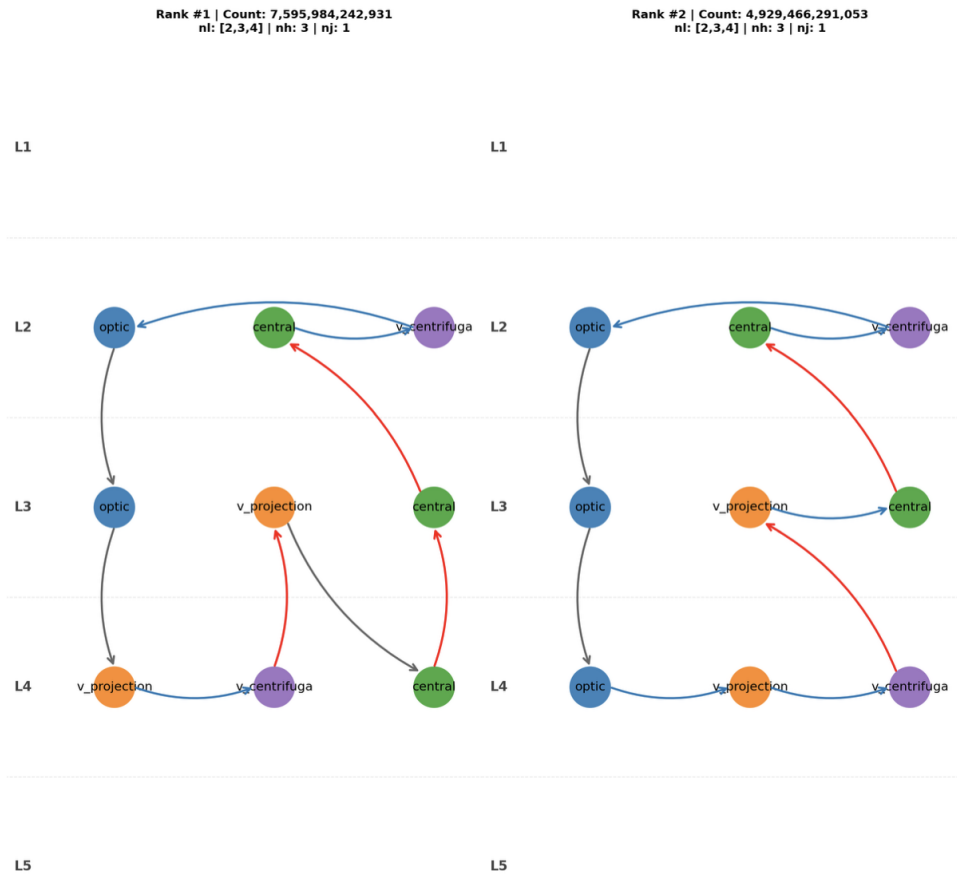
**Fig. 12** Top-3 ranked motifs discovered with parameters  $NL = \{2, 3, 4, 5\}$ ,  $NH = 2$ ,  $NJ = 1$ .



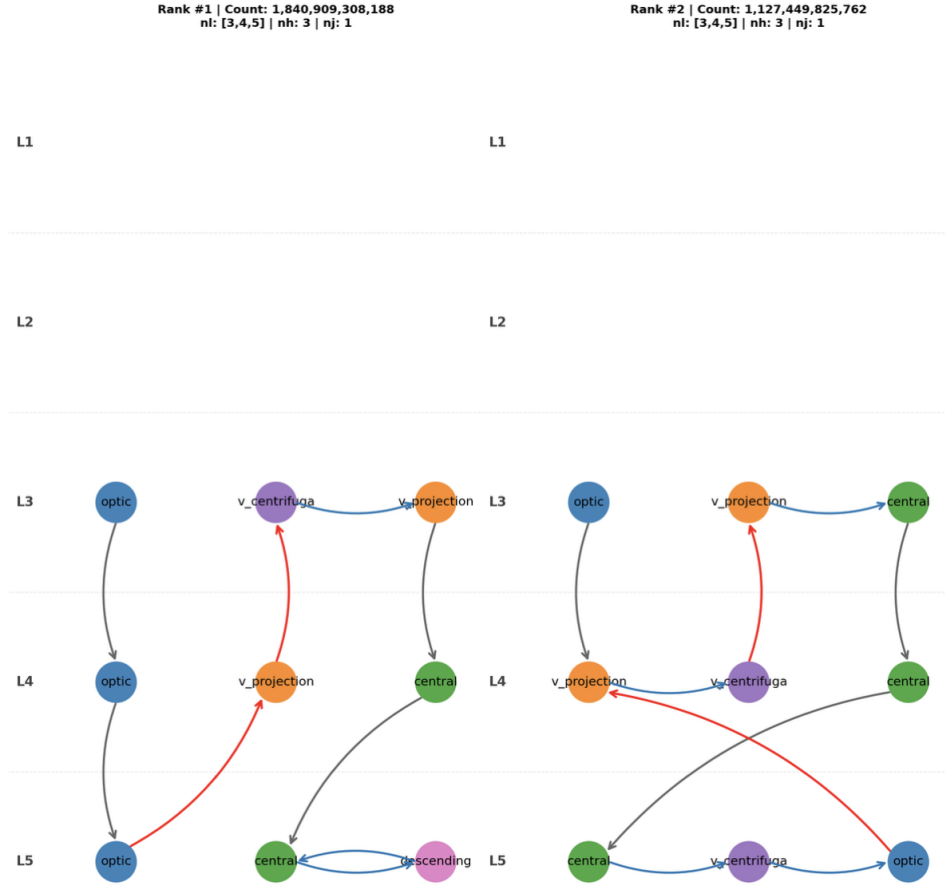
**Fig. 13** Top-3 ranked motifs discovered with parameters  $NL = \{1, 2, 3, 4, 5\}$ ,  $NH = 2$ ,  $NJ = 1$ .



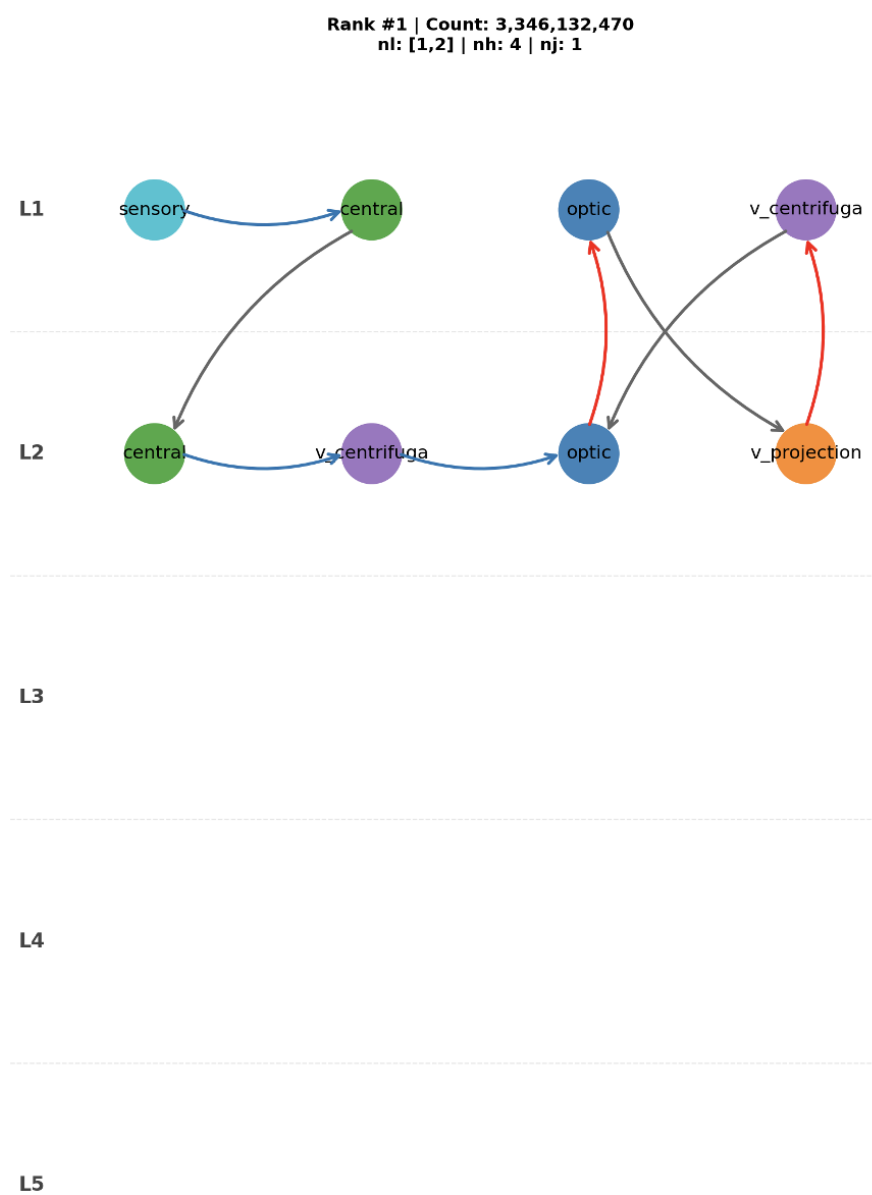
**Fig. 14** Top-2 ranked motifs discovered with parameters  $NL = \{1, 2, 3\}$ ,  $NH = 3$ ,  $NJ = 1$ .



**Fig. 15** Top-2 ranked motifs discovered with parameters  $NL = \{2, 3, 4\}$ ,  $NH = 3$ ,  $NJ = 1$ .



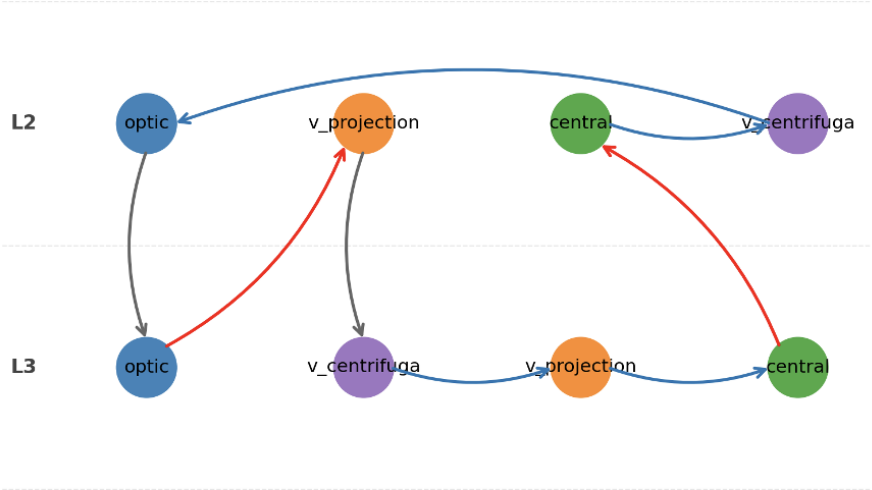
**Fig. 16** Top-2 ranked motifs discovered with parameters  $NL = \{3, 4, 5\}$ ,  $NH = 3$ ,  $NJ = 1$ .



**Fig. 17** Top-1 ranked motifs discovered with parameters  $NL = \{1, 2\}$ ,  $NH = 4$ ,  $NJ = 1$ .

Rank #1 | Count: 25,260,816,462  
nl: [2,3] | nh: 4 | nj: 1

L1



L4

L5

**Fig. 18** Top-1 ranked motifs discovered with parameters  $NL = \{2, 3\}$ ,  $NH = 4$ ,  $NJ = 1$ .

Rank #1 | Count: 234,073,267,800  
nl: [3,4] | nh: 4 | nj: 1

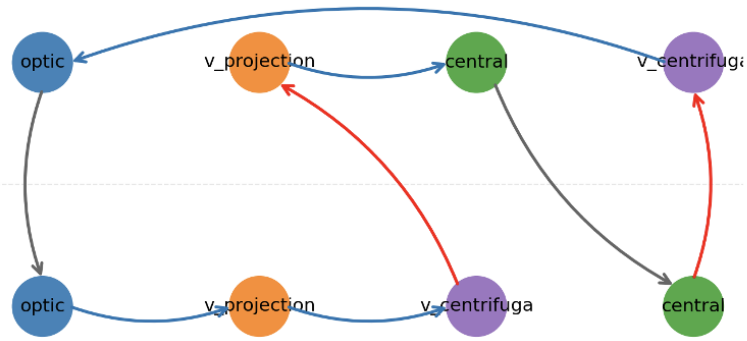
L1

L2

L3

L4

L5



**Fig. 19** Top-1 ranked motifs discovered with parameters  $NL = \{3, 4\}$ ,  $NH = 4$ ,  $NJ = 1$ .

Rank #1 | Count: 29,400,209,219  
nl: [4,5] | nh: 4 | nj: 1

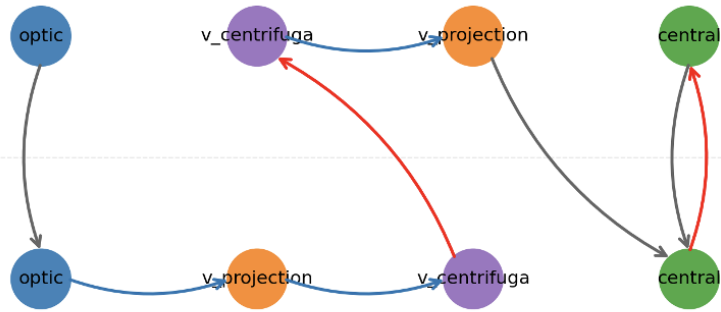
L1

L2

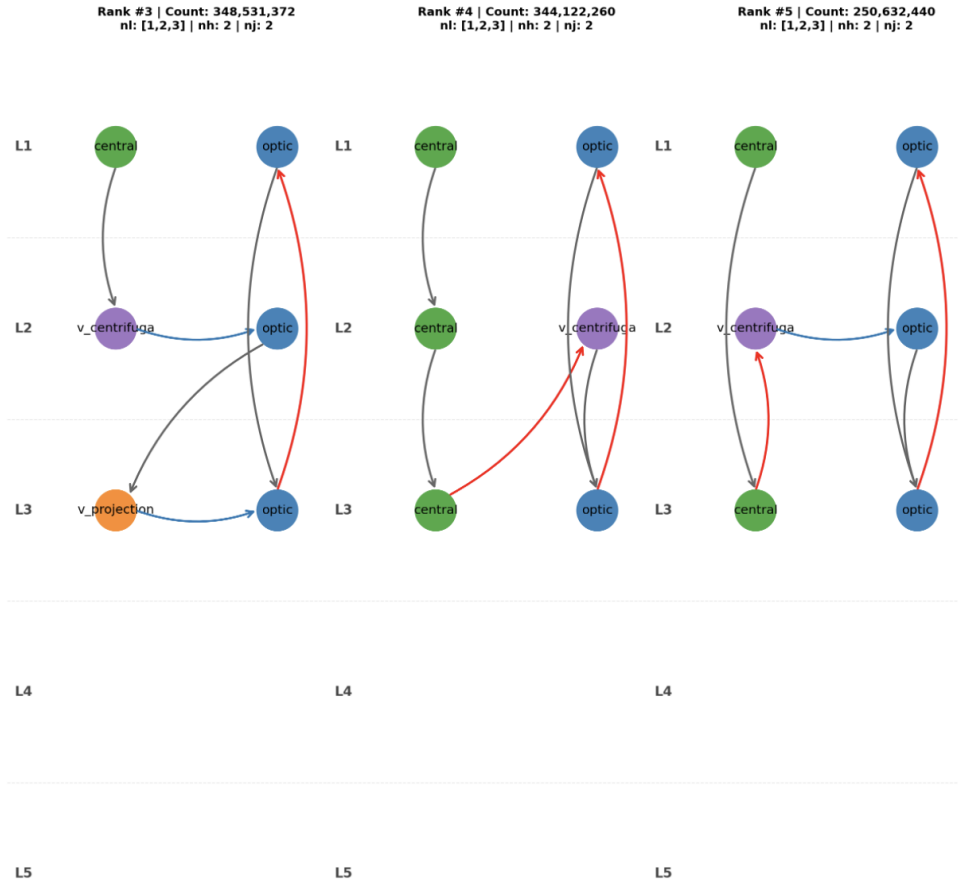
L3

L4

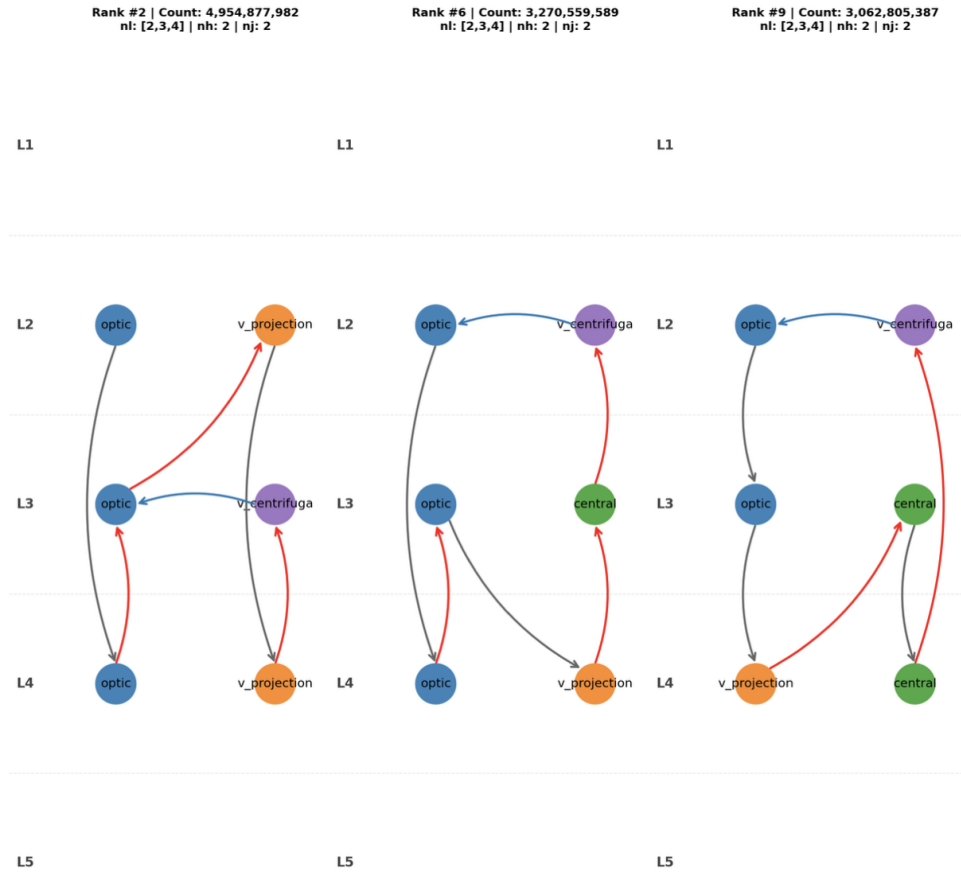
L5



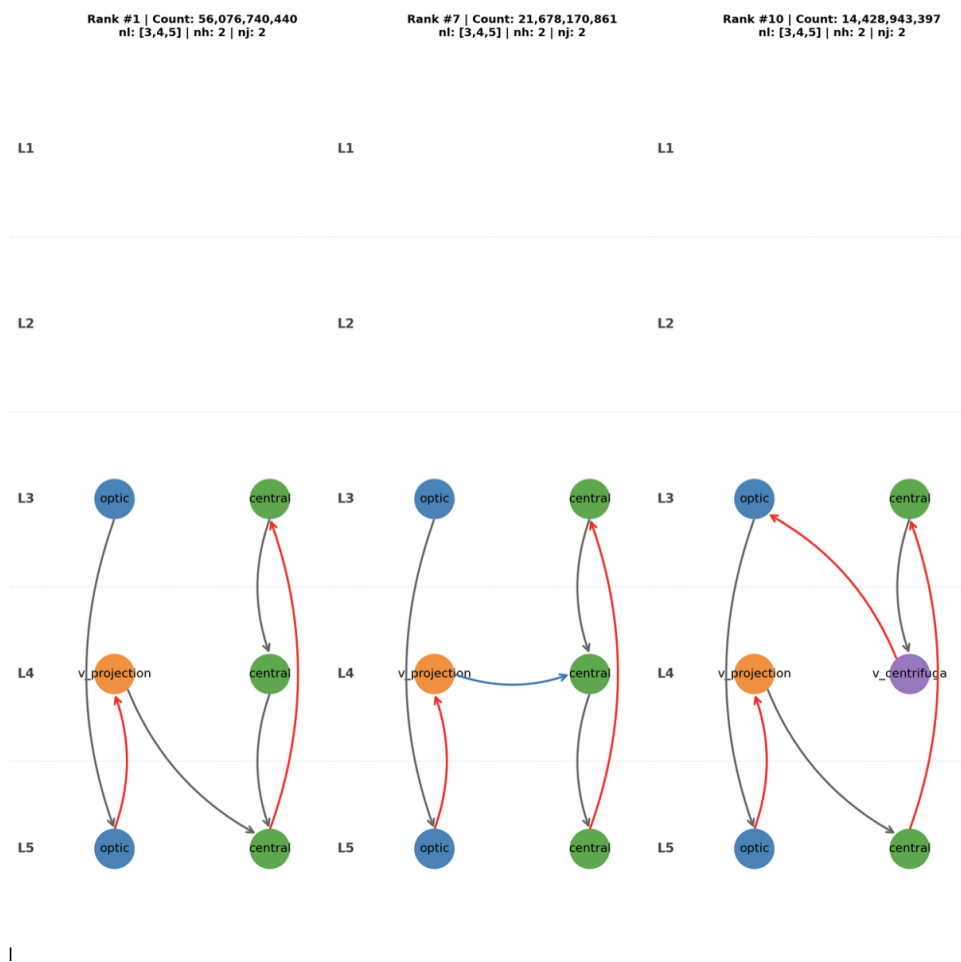
**Fig. 20** Top-1 ranked motifs discovered with parameters  $NL = \{4, 5\}$ ,  $NH = 4$ ,  $NJ = 1$ .



**Fig. 21** Selected ranked motifs discovered with parameters  $NL = \{1, 2, 3\}$ ,  $NH = 2$ ,  $NJ = 2$ .

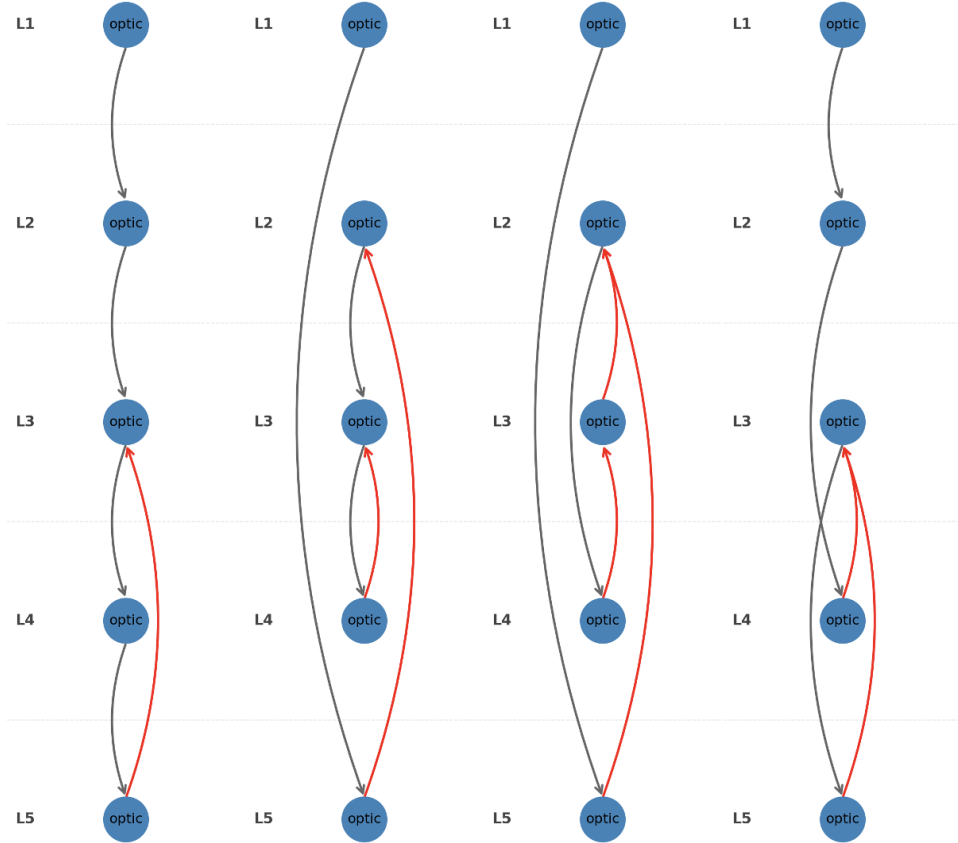


**Fig. 22** Selected ranked motifs discovered with parameters  $NL = \{2, 3, 4\}$ ,  $NH = 2$ ,  $NJ = 2$ .



**Fig. 23** Selected ranked motifs discovered with parameters  $NL = \{3, 4, 5\}$ ,  $NH = 2$ ,  $NJ = 2$ .

Rank #1 | Count: 1,037,896,021 | ni: [1,2,3,4,5] | nh: 1 | nj: 4      Rank #3 | Count: 801,080,707 | ni: [1,2,3,4,5] | nh: 1 | nj: 4      Rank #4 | Count: 771,220,982 | ni: [1,2,3,4,5] | nh: 1 | nj: 4      Rank #8 | Count: 595,696,226 | ni: [1,2,3,4,5] | nh: 1 | nj: 4



**Fig. 24** Selected ranked motifs discovered with parameters  $NL = \{1, 2, 3, 4, 5\}$ ,  $NH = 1$ ,  $NJ = 4$ .